

OCEANS OF DATA



Tasha Gownaris
Gettysburg College

Oceans of Data

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 07/21/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

TABLE OF CONTENTS

Licensing

1: Our Oceans and Oceanographic Data

- 1.1: Overview of the Oceans
- 1.2: Continental Margins
- 1.3: Marine Provinces
- 1.4: How are data about the ocean collected?
- 1.5: How are oceanographic data displayed?

2: Plate Tectonics and Ocean Bathymetry

- 2.1: Spatial Data in the Oceans- Bathymetry, Latitude, and Longitude
- 2.2: A Brief Overview of Earth's Structure
- 2.3: Alfred Wegener and the Theory of Plate Tectonics
- 2.4: Types of Plate Boundaries and the Features they Create
- 2.5: Seamounts, Hot Spots, and Hydrothermal Vents
- 2.6: Practice- Identify the Ocean Features!
- 2.7: Earthquakes and Plate Tectonics

3: Ocean Sediments

- 3.1: Classifying Sediments
- 3.2: Lithogenous Sediments
- 3.3: Biogenous Sediments
- 3.4: Hydrogenous Sediments
- 3.5: Cosmogenous Sediments
- 3.6: Sediment Distribution

4: Salinity, Temperature, and Density in the Ocean

- 4.1: Properties of Water
- 4.2: Ocean Salinity
- 4.3: Ocean Temperature
- 4.4: Ocean Density and Pressure

5: CO₂, O₂, and the Ocean

- 5.1: Measuring Past and Current CO₂ Concentrations
- 5.2: Biological and Physical Processes Impacting CO₂
- 5.3: Profiles of Carbon Dioxide, Oxygen, and Nutrients
- 5.4: CO₂ and Ocean Acidification
- 5.5: Earth's Heat Budget
- 5.6: Climate Change

6: Circulation in the Atmosphere and Ocean

- 6.1: Winds and the Coriolis Effect
- 6.2: Winds and Climate
- 6.3: Surface Gyres
- 6.4: The Gulf Stream

- [6.5: Currents, Upwelling and Downwelling](#)
- [6.6: El Niño and La Niña](#)
- [6.7: Thermohaline Circulation](#)

[7: Plankton, Primary Production, and Hypoxia](#)

- [7.1: Primary Production & Primary Producers](#)
- [7.2: Drivers of & Patterns in Primary Production](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Our Oceans and Oceanographic Data

Learning Objectives

After reading this chapter, you should be able to:

- describe the five major ocean basins, the major ocean provinces, and the characteristics that differentiate/define them
- recognize several of the common sensors and platforms used to collect oceanographic data
- identify key components of a scientific graph (axes scales, legends, multiple y-axes)
- describe data patterns such as trends over time or distance
- recognize graph types commonly used in oceanography (time series, contour plots, vertical sections, depth profiles)

[1.1: Overview of the Oceans](#)

[1.2: Continental Margins](#)

[1.3: Marine Provinces](#)

[1.4: How are data about the ocean collected?](#)

[1.5: How are oceanographic data displayed?](#)

This page titled [1: Our Oceans and Oceanographic Data](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by [Tasha Gownaris](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

1.1: Overview of the Oceans

Let's begin by looking at a few basic facts about the oceans. We often think of Earth in terms of its land area, but in reality 71% of the Earth's surface is covered by oceans, while only 29% is land. Oceans cover an area of 139 million miles² or 361 million km², and contain a volume of about 1.37 billion km³ of water. All of this water is not distributed equally over the Earth; 61% of the Northern Hemisphere is covered by oceans, while in the Southern Hemisphere the oceans cover 81% of the surface area (Figure 1.1.1).

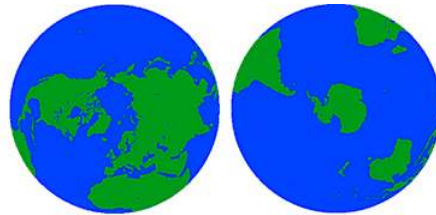


Figure 1.1.1 Ocean cover in the Northern (left) and Southern (right) Hemispheres.

Various sources differ in the number of recognized ocean basins. Historically the major oceans were recognized as the Pacific, Atlantic, Indian, and Arctic Oceans. More recently, the Southern Ocean has been recognized as fifth named ocean, comprising all of the water from the coast of Antarctica to 60° S (Figure 1.1.2). In 2000 these boundaries were submitted to the [International Hydrographic Organization](#) for official recognition, but several countries do not recognize it as a separate ocean, but rather as the southern extension of the other major oceans. The Southern Ocean has its own unique characteristics, so for the purposes of this book we will include it as a separate ocean.

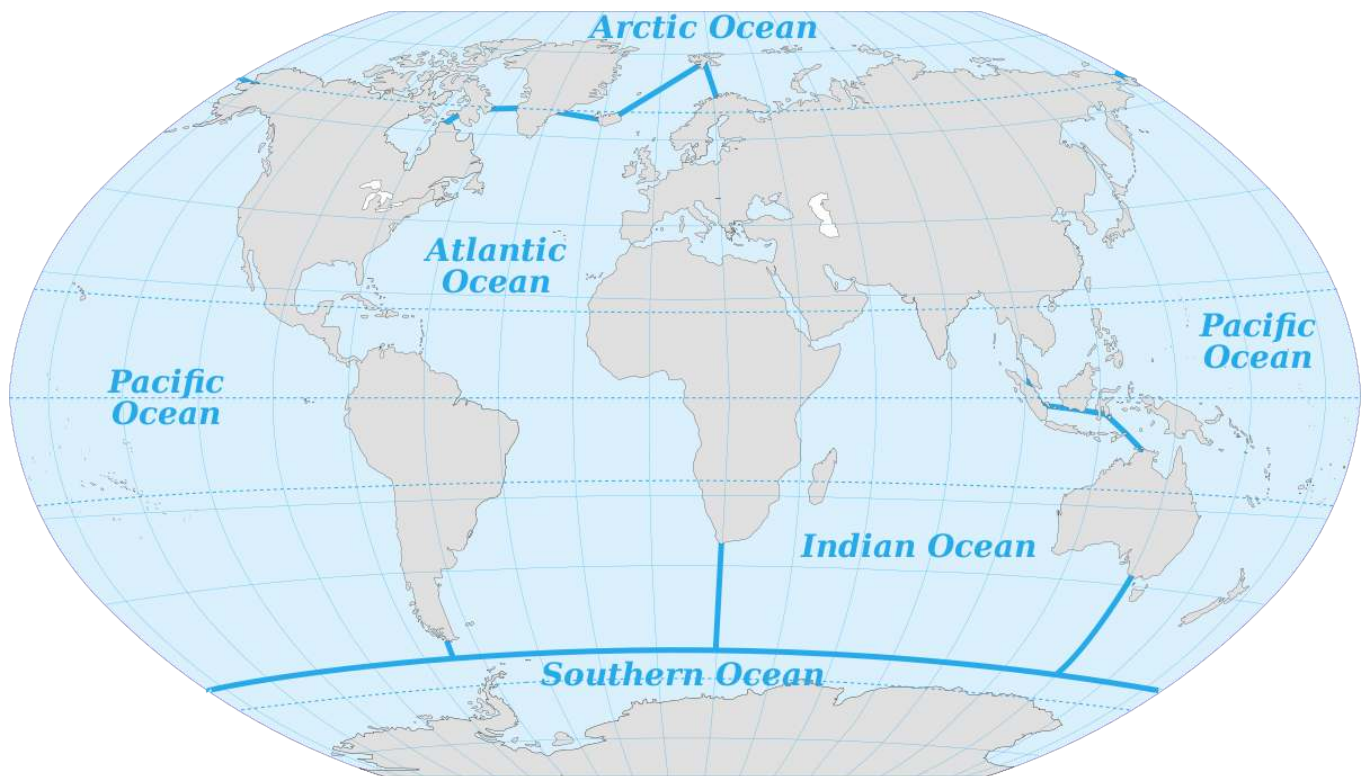


Figure 1.1.2 Map of the world oceans (By Pinpin [GFDL (<http://www.gnu.org/copyleft/fdl.html>) or CC BY-SA 3.0], via Wikimedia Commons).

The oceans account for vast amounts of water, containing 97% of the water on Earth's surface, with over half of the water in the Pacific alone (Table 1.1.1).

Table 1.1.1 Percentage of Earth's water in various locations

Pacific	52%
---------	-----

Atlantic	25%
Indian	20%
Ice	2%
Ground water	0.6%
Atmosphere, lakes & rivers	0.01%

The average depth of the world ocean is about 3800m (12,500 ft), which is about four times deeper than the average land elevation is high (840m or 2800 ft). In fact Mt. Everest, the highest point on land, is 8848m (29,028 ft) high, while the deepest part of the ocean, the Challenger Deep of the Marianas Trench is 11,022m (36,200 ft) deep. So you could submerge Mt. Everest in the Marianas Trench and it would still be covered by over 2 km of water! Because there is so much more water on Earth than there is land, if you could smooth out the land elevation the entire Earth would still be covered by water about 2700 m deep.

Of the major ocean basins, the Pacific is the largest (almost as large as all of the others combined), and is the deepest (Table 1.1.2).

Table 1.1.2 Area and depth of the major oceans

	Area (million km ²)	Average depth (m)
Pacific	166	4282
Atlantic	87	3926
Indian	73	3963
Arctic	14	1205
Southern	20	4000

Watch the video below for some perspective on the size and depth of the oceans.



This page titled [1.1: Overview of the Oceans](#) is shared under a [CC BY 4.0](#) license and was authored, remixed, and/or curated by [Tasha Gownaris](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

- [1.1: Overview of the Oceans](#) by [Paul Webb](#) is licensed [CC BY 4.0](#). Original source: <https://rwu.pressbooks.pub/webboceanography>.

1.2: Continental Margins

Continental margins refer to the region of transition from the land to the deep seafloor, i.e. between continental and oceanic crust. Figure 1.2.1 shows an idealized continental margin. When examining this figure, and others like it, note that there is significant vertical exaggeration; the depth scale covers approximately 5000 m, while the horizontal scale extends around 300 km. This makes the features look much steeper than they actually are. The bar at the bottom of Figure 1.2.1 shows what the margin would look like without this exaggeration; there is a much more gradual transition to depth.

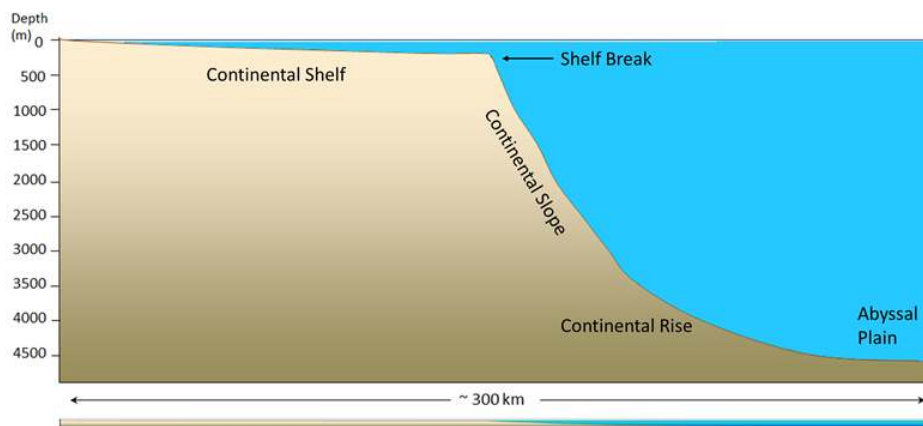


Figure 1.2.1 Features of a passive continental margin (modified by PW from Steven Earle, "Physical Geology").

The **continental shelf** is the shallow, flooded edge of the continent. Geologically the shelf is still part of the continental crust, but it is often overlaid with marine sediments. On average, the shelf extends about 80 km from the coast; some margins have very little shelf, while the Siberian Shelf in the Arctic extends roughly 1500 km. The depth of the shelf generally remains below about 150 m, and the floor of the shelf is fairly flat. The flat topography is the result of changes in sea level; throughout history the shelves have been both submerged and exposed, and as sea level rose and fell, wave action, ice sheets, and other erosional processes smoothed out the shelf surface. Wave action and the movement of sediments over the shelf have continued this smoothing process. Continental shelves only make up about 6% of the ocean's surface area, but they are biologically one of the richest parts of the ocean; their shallow depth prevents nutrients from sinking out, and their proximity to the coast provides significant nutrient input. The continental shelf ends at the **shelf break**, which is the point where the angle of the seafloor begins to get steeper. The shelf break averages about 135 m deep.

After the shelf break, the seafloor takes on a steeper angle (about 4°) as it descends to the deep ocean. This steeper portion of the margin is the **continental slope**, and it extends from the shelf break down to 3000-5000m. In some parts of the ocean, large submarine canyons have been carved into the continental slope; for example, Monterey Canyon in Monterey Bay, California, is a submarine canyon similar in size to the Grand Canyon! These canyons may be carved out by **turbidity currents**, which are essentially landslides of sediment, rocks, and other debris down the face of the slope.

At the bottom of the slope is the **continental rise**. This area represents where the continental crust meets the oceanic crust, as the slope begins to level off to become the deep ocean floor. The rise consists of a thick layer of accumulated sediment coming from the continent, so it is difficult to tell where the slope ends and the rise begins.

After the rise comes the **abyssal plain**, or the deep ocean floor, lying between 4500 – 6000 m. The abyssal plain includes most of the ocean floor, and is the flattest region on Earth. It is flat due to millions of years of sediment accumulation on the bottom, which buries many bottom features (Figure 1.2.2).

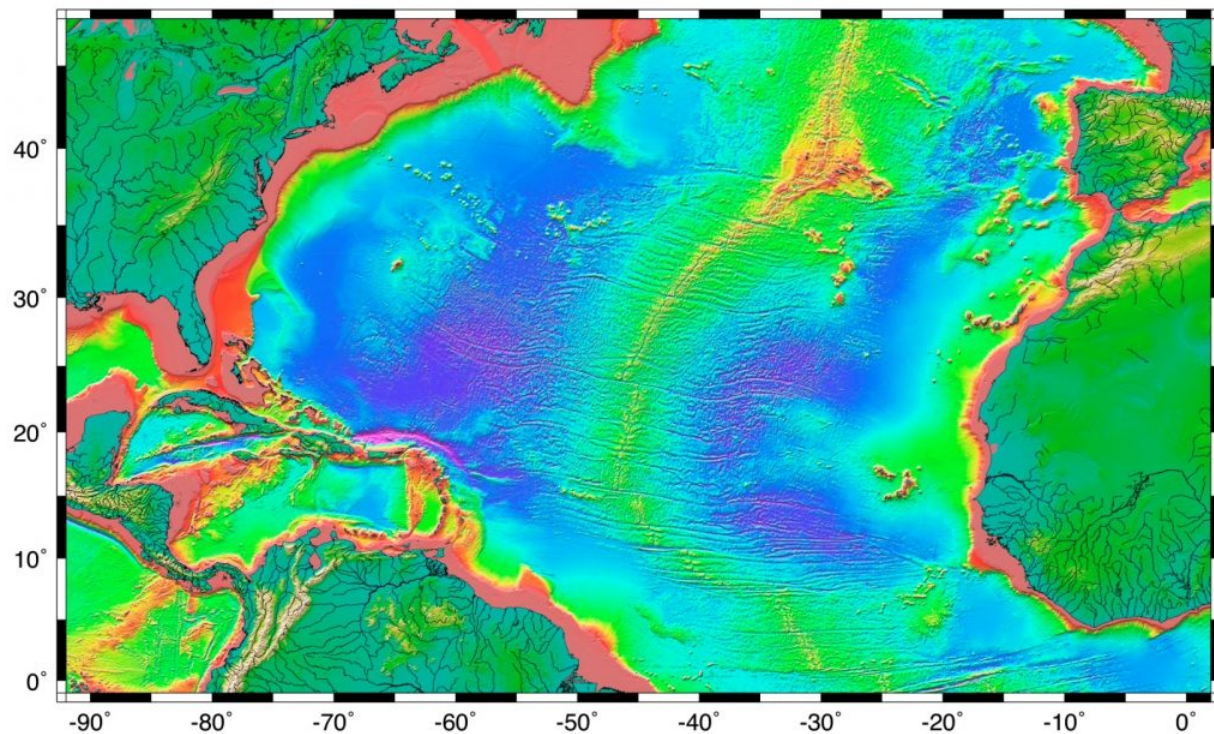


Figure 1.2.2 Topography of the North Atlantic. Shallow continental shelf regions are shown in red, and the abyssal plain is shown in blue. Along the east coast of the United States the continental slope can be seen in green (Steven Earle, “Physical Geology”).

* “Physical Geology” by Steven Earle used under a CC-BY 4.0 international license. Download this book for free at <http://open.bccampus.ca>

This page titled [1.2: Continental Margins](#) is shared under a [CC BY 4.0](#) license and was authored, remixed, and/or curated by [Tasha Gownaris](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

- [1.2: Continental Margins](#) by [Paul Webb](#) is licensed [CC BY 4.0](#). Original source: <https://rwu.pressbooks.pub/webboceanography>.

1.3: Marine Provinces

In section 1.2 we learned about the regions that make up the continental margins. So before we leave this topic, we will look at some of the other ways we can categorize the ocean environments.

The first major distinction is between the pelagic and benthic zones. The **pelagic** zone refers to the water column, where swimming and floating organisms live. The **benthic** zone refers to the bottom, and organisms living on and in the bottom are known as the **benthos**.

The pelagic zone is divided into two provinces: the **neritic** province corresponds to all of the water from the low tide line to the shelf break, while the **oceanic** province represents all of the other water in the open ocean regions.

The oceanic province is divided into depth zones (Figure 1.3.1):

- 0-200 m is the **epipelagic** zone (“epi” = “upon”, as in on top of the pelagic zone). This is the region where enough light penetrates the water to support photosynthesis (see section 7.3), so it is also called the **euphotic** or **photic** zone.
- 200-1000 m is the **mesopelagic** zone (“meso” = “middle”). There is some light here, but not enough for photosynthesis, so it is called the **dysphotic** zone, or the **twilight zone**.
- 1000-4000 m is the **bathypelagic** zone (“bathy” = “deep”). There is no light at these depths, so it is referred to as the **aphotic** zone. About 75% of the living space in the ocean lies at these depths or deeper.
- 4000-6000 m is the **abyssopelagic** or **abyssalpelagic** zone, which extends to the seafloor in most areas.
- 6000 m and below is the **hadopelagic** or **hadalpelagic** zone (named for Hades or “hell”). This refers to the water in deep ocean trenches.

Inhabitants of these regions are referred to according to their habitat, for example mesopelagic fish, or epipelagic squid.

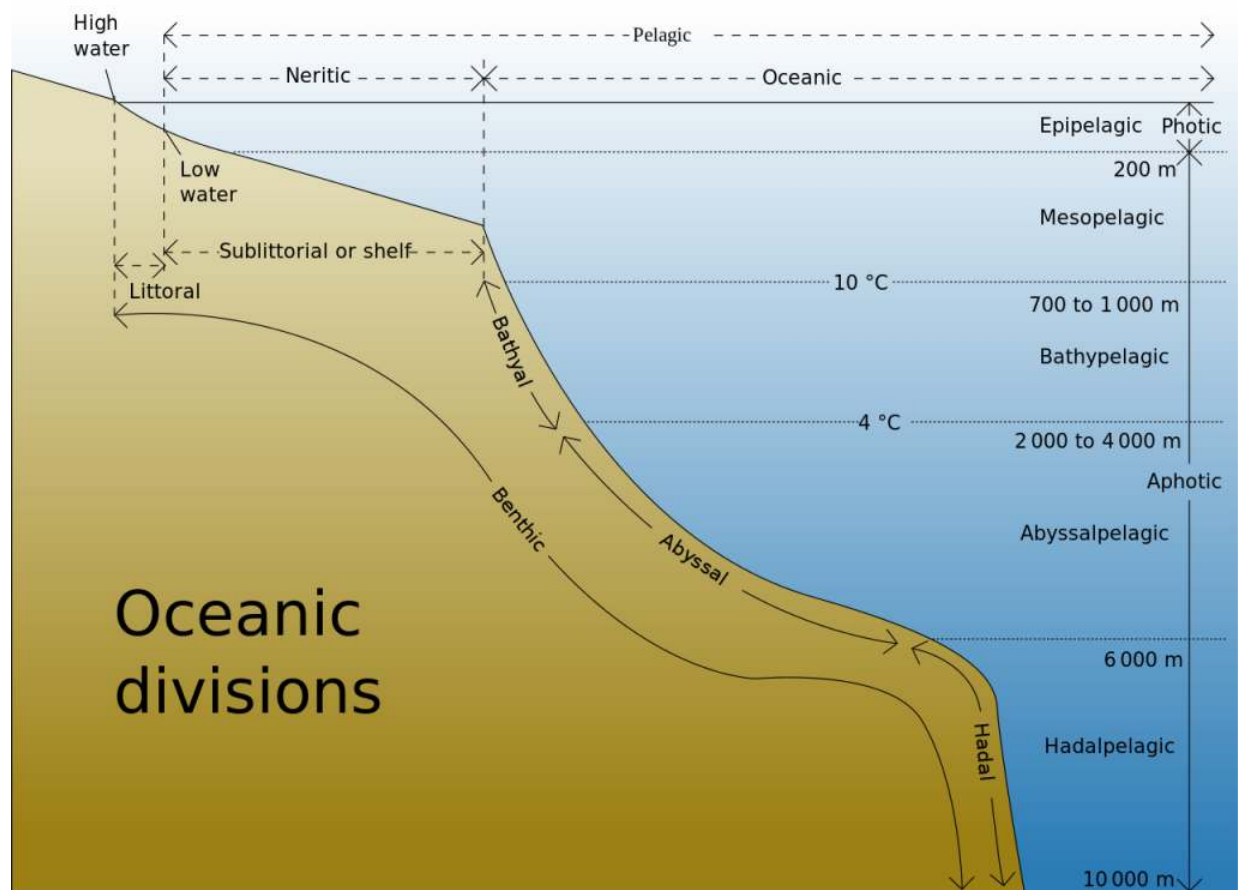


Figure 1.3.1 The major benthic and pelagic oceanic divisions (K. Aainsqatsi at en.Wikipedia [Public domain, GFDL (<http://www.gnu.org/copyleft/fdl.html>) or CC-BY-SA-3.0], via Wikimedia Commons).

The benthic environment is also divided into zones, most of which correspond to the pelagic divisions:

- The **supralittoral** zone lies above the high tide line. Also called the spray zone, it is only submerged during storms or unusually high waves.
- The **littoral** zone is the region between the high and low tides. Thus it is also referred to as the **intertidal** zone.
- Below the littoral zone is the **sublittoral** (shelf) zone, extending from the low tide mark to the shelf break, essentially covering the continental shelf.
- The **bathyal** zone extends along the bottom from the shelf break to 4000m, so it generally includes the continental slope and rise.
- The **abyssal** zone is found between 4000-6000 m, including most of the abyssal plains. The abyssal zone represents about 80% of the benthic environment.
- The **hadal** zone includes all benthic regions deeper than 6000 m, such as in the bottom of trenches.

This page titled [1.3: Marine Provinces](#) is shared under a [CC BY 4.0](#) license and was authored, remixed, and/or curated by [Tasha Gownaris](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

- [1.3: Marine Provinces](#) by [Paul Webb](#) is licensed [CC BY 4.0](#). Original source: <https://rwu.pressbooks.pub/webboceanography>.